STATISTICS

Chapter 9 Homework

Hypothesis testing: conceptual overview

Chapter 9 conceptual coverage only | Estimated time: about 1 hour

Complete all problems and show your work on separate paper. Plan for about 45-50 minutes of work and 10-15 minutes to check, correct, and submit.

Use available resources to check your answers and reasoning. Complete the self-evaluation in Blackboard, upload a clear photo of your one best representative page, and retain all your written work.

A four-function calculator is sufficient; needed advanced numerical values are supplied. Show probability setups, units, and brief explanations. Round probabilities to four decimals unless exact fractions are convenient. All scenarios and data are fictional.

Claims, evidence, and decisions

1. A shipping service claims mean delivery time is 3 days. A study investigates whether the population mean exceeds 3 days. Define the population parameter and write the null and alternative hypotheses.

2. For Problem 1, a valid test returns p = 0.03. Explain this p-value in terms of the null model and results at least as extreme as observed, in the direction of the alternative. Why is it not a 3% probability that the null hypothesis is true?

3. Using p = 0.03, make a decision at (a) significance level 0.05 and (b) 0.01. Write a conclusion about the mean delivery time for each level.

4. Another study returns p = 0.40 for the same hypotheses. Write an appropriate conclusion. Explain why “the mean is proved to equal 3 days” is too strong.

Errors and the quality of evidence

5. Describe a Type I error and a Type II error for Problem 1. Give one practical consequence of each.

6. A team lowers its significance level from 0.05 to 0.01 with its sample size and test otherwise unchanged. Explain the effect on Type I error risk and the general tradeoff with power.

7. A large study finds an estimated increase of 0.01 day and p = 0.001. What does the p-value say? What additional information helps judge practical importance?

8. A valid two-sided test of a population mean uses the same assumptions and method as a 95% confidence interval. The interval is (3.2, 3.8) days. At significance level 0.05, what is the decision for a null mean of 3? Explain without calculating a test statistic.

9. A researcher tries 20 outcome measures and reports only the smallest p-value. Explain why this selective reporting can mislead. What should be disclosed or planned before data analysis?

10. A report says: “p = 0.02, so there is a 98% chance our claim is true, the effect is large, and it applies to everyone.” Identify and correct all three unsupported conclusions.